

A Reconfigurable Mobile Head-Mounted Display Supporting Real World Interactions

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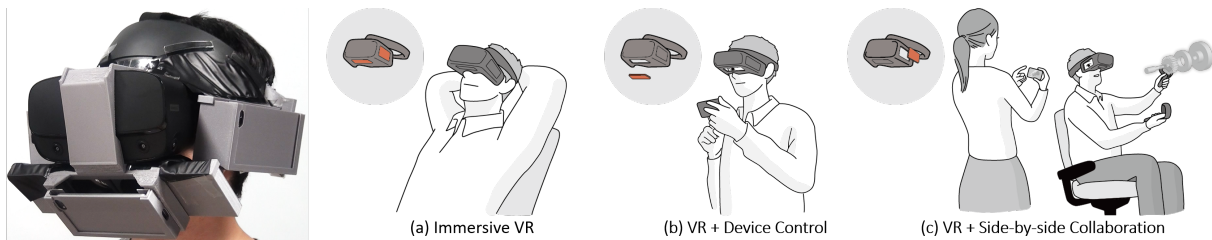


Figure 1: ModularHMD's main use cases: (a) a user is fully immersed in VR content using our HMD with three modules displaying extended VR views; (b) the user then removes the bottom module and uses it as an input device to interact with the VR content; (c) the user can also remove the side module to ensure peripheral awareness of real-world partners during a VR experience and use the module as an instant interaction device for a partner, allowing quick collaboration.

ABSTRACT

We propose a new mobile head-mounted display, ModularHMD, that uses a modular mechanism with a manually reconfigurable structure to enable ad-hoc peripheral interaction with real-world objects and people while maintaining an immersive VR experience. We designed and built a proof-of-concept prototype of ModularHMD using a base commercial HMD and three removable display modules installed in the periphery of the HMD cowl. The user can rapidly switch between different HMD configurations according to their needs. The modules can be removed to ensure peripheral awareness of the real world, used as instant interaction devices (e.g., keyboards), and then returned to their original positions to terminate the peripheral interaction.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; *Mixed / augmented reality*.

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CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

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ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451765>

KEYWORDS

Virtual Reality, Head-mounted display, Real-world interaction, Asymmetric interaction

ACM Reference Format:

Isamu Endo, Kazuki Takashima, Maakito Inoue, Kazuyuki Fujita, Kiyoshi Kiyokawa, and Yoshifumi Kitamura. 2021. A Reconfigurable Mobile Head-Mounted Display Supporting Real World Interactions. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3411763.3451765>

1 INTRODUCTION

Head-mounted displays (HMDs) have become more affordable and portable, no longer requiring external equipment. Such mobile HMD setups have huge potential beyond entertainment use [30], enabling mobile work contexts [7] in which users can enter virtual workspace while they are physically in office [11, 24], in shared spaces [32], or on transportation [1, 19], etc. In such mobile VR scenarios, an HMD user would require a certain level of situational awareness of the real world to respond to sudden physical interactions or nearby actions from non-HMD users. However, current HMDs use occlusive cowls to induce an immersive VR experience, which fully isolates the HMD user [14] from the real world. We call this isolation the “HMD Boundary” and it causes two difficulties for the user. First, HMD users have no awareness or control with real-world objects (e.g., keyboards) without taking off the

headset [3, 8, 27]. Second, HMD users have limited spatial awareness [17] and are not capable of collaboratively interacting with both the VR content and with nearby people in mobile work contexts because external shared devices are not always available. A coherent solution is needed to address these issues altogether for successful mobile HMDs.

Many researchers have worked on these two issues in separate work. For the issue of object interaction, a common approach is the use of video-see-through (VST) that visualizes surrounding objects in the VR view (e.g. [3]). Another approach is redesigning the HMD. Multi-channel dynamic immersion (MDI) [27] allows a user's peripheral viewing of real-world devices through transparency-controllable LCD panels. For the issue of ad-hoc collaboration, the challenges become more complicated because awareness from both users' perspectives should be ensured. FaceDisplay [10] deals with this issue using three outward-facing touchscreens affixed on the HMD, allowing co-located collaboration between users through touch interactions around the HMD user's head. These shows that many researchers have recognized HMD boundary issues and proposed solutions for them. However, no currently usable mobile HMD has yet been developed to simultaneously deal with the two types of peripheral interactions with real-world objects and partners.

Inspired by previously suggested HMD concepts (e.g., [4, 10, 16, 20, 22, 27]), we focus on HMD form and aim to expand the idea by introducing a reconfigurable structure that can switch between different functions. Thus, we propose a new reconfigurable HMD concept using a modular mechanism to support ad-hoc peripheral interaction with real-world objects and partners while maintaining an immersive VR experience for the HMD user. The main concept is the installation of detachable modular mechanisms around the periphery of an HMD, allowing the user to switch between different views and functions by removing these modules to ensure real-world awareness, reusing the removed modules as instant hand-held devices to facilitate subsequent interaction, and returning the modules to end peripheral interaction. Fig. 1 illustrates our concept, in which we configured a regular HMD along with three removable display modules for its side and lower parts. The basic configuration with all modules attached offers an extended field of view (FOV) via inward-facing side and bottom display modules (Fig. 1(a)). This HMD allows the user to remove the side and/or lower modules to allow peripheral viewing of the real world (Fig. 1(b)(c)). This HMD also allows the removed modules to be used as interaction devices (e.g., touchscreens) by the HMD users themselves or by real-world partners. Such cyclic interaction would allow the HMD setup to remain fully mobile and self-contained.

2 RELATED WORK

2.1 HMD boundary issues

Grubert et al. [7] argued that high mobility and immersiveness are core elements of HMDs for location-independent offices of the future that use virtual reality technology. They summarized critical challenges of existing HMDs, including lower situational awareness of the real world, poor physical interaction, hindered communication among users, and poor text typing performance.

These have been widely recognized by VR users and researchers (e.g., [10, 17, 27]).

2.2 HMD user situational awareness for physical interaction

Mobile HMDs will be used in shared spaces [14, 32], homes [29], offices [11, 24], and in transportation [1, 19]. In each situation an HMD user requires a certain level of situational awareness of the real world. The simplest solution is to take off the HMD headset. However, HMD users can see either the virtual environment (VE) or the real world, but not simultaneously. Non-cowl HMD (e.g., [28]) might be useful to make peripheral awareness always available but it obviously inhibits immersive experiences. A common solution is to overlay captured real-world objects and people on the VE, which includes the Augmented Virtuality (AV) approach [3, 12, 17, 20, 29, 29], and displaying real-world people or objects as previously prepared 3D models [5, 8, 26]. While these visual cues increase the situational awareness of HMD users, they do not help bystanders or facilitate collaborative interactions between them.

A closely related approach to ours is transparency-controllable HMD that allows HMD users to see the real world (e.g. Compound-Dome [16], MDI [27]). MDI is a dynamic immersion-controllable HMD that uses multiple optical channels. Transparency-controllable LCD panels are installed in the cowl of a HMD, enabling users peripheral vision. A MDI user can see and access peripheral devices (e.g., smartphones, keyboards, mouse) through LCD panels without taking off the headset. We were inspired by MDI and we significantly expanded this idea in two ways: enabling direct vision to the real world and cross-device collaboration between HMD and non-HMD users.

2.3 Interaction between HMD and non-HMD users

To present VE content to nearby users, there are many previous works using external devices such as desktop displays, touch displays [12], floor projections [9], etc. However, these solutions require additional setup in advance and might not be suited for ad-hoc collaborative interaction or mobile VR scenarios. To enable asymmetric interaction between HMD and non-HMD users, many specially designed HMDs with monitor(s) attached to their faces have been proposed [4, 10, 15, 21]. FaceDisplay [10] has three touch-sensitive displays and a depth camera at the back of the HMD, allowing asymmetric collaboration between HMD and non-HMD users through touch or gesture interactions. The idea is related to our work, but FaceDisplay is primarily intended for entertainment applications with unique physical formations and interactions between users. Our approach is meant to provide distributed workspaces for HMD and non-HMD users, where more natural formations (e.g. side-by-side) can be made.

2.4 Modular mobile devices for HMD

To mitigate HMD boundary issues, we pay particular attention to shape-changing interfaces that enables dynamic forms of interface according to a given situation. Within this context, a modular device (e.g., [6]) might be promising for HMD design. Similar concepts have been verified on FOV extension [22], flexible AR functions [20],

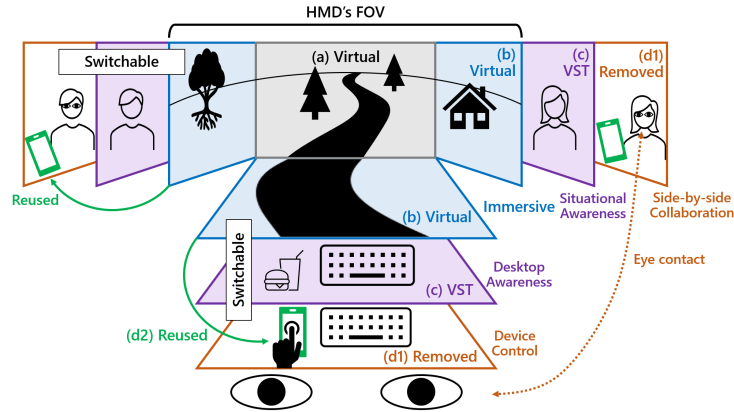


Figure 2: All configuration categories of ModularHMD by different states of three modules.

and achieving various VR interaction [18] but this idea has not been well studied.

Another unique feature of a modular device concept is reusing free modules for different purposes. An example is “device lending” [25], where the owner of the main modular smartphone can give interaction authority to a trusted person by giving them one of the modules. Once the instant interaction is over, the module and interaction authority can be returned to the owner by returning the module. This approach is efficient and straightforward to manage the authentication process, as compared to ad-hoc use of their own devices. We try to incorporate this unique feature into our HMD design.

3 MODULAR HMD

We propose ModularHMD, a new HMD concept allowing HMD users to enable peripheral interaction with real-world objects and partners by manually reconfiguring the HMD structure. We focused on detachable feature in our modular device concept that tackles HMD boundary issues.

To design our new HMD concept using modular mechanics, we had three goals: 1) an immersive VR experience is the main priority, 2) each module should support ad-hoc peripheral interaction with the real world, and 3) the HMD should be sufficiently mobile and all functions should be self-contained. These goals make our design space slightly narrower but allow us to focus on how peripheral modules can be used for sub-interactions while maintaining an immersive VR experience. In the rest of the section, we explain the three display modules at the sides and bottom of the HMD.

3.1 HMD Configurations

Fig. 2 is a structural overview of ModularHMD comprising one frontal main display, two side display modules, and one bottom display module. The three modules are easily detachable. This figure illustrates ModularHMD’s main configuration categories that allow users to work on different tasks. The central rectangular area in gray shown in Fig. 2 (a) indicates the basic (commercially available) HMD that enables a high-quality VR experience to support the primary tasks. We add two side modules and one bottom module, while each can be switched among four states: no display content,

VR extension view (in blue in Fig. 2 (b)), VST view (purple (c)), and removed/reused (orange (d1) and green (d2)). Depending on the desired peripheral interaction, each module can be manually switched. The side and bottom modules are designed to ensure desktop visions and spatial awareness respectively when desired.

3.2 Interaction designs using different HMD Configurations

If all of the display modules are turned off and attached to the HMD, the user would feel like wearing a regular occlusive HMD while being fully isolated from the real world. This configuration is trivial as it is identical to the base functionality of the HMD. The following paragraphs explain the main interaction patterns for certain particular configurations of ModularHMD and our software working across HMD and modules, as illustrated in Fig. 3.

(a) Immersive VR Experience (All modules: VR extension):

If all of the modules are attached and display the extended views of the VR content, the user would have a more immersive VR experience augmented by the peripheral VR views [13, 22, 31] provided by the inner-facing display modules. Fig. 3(a) shows a 360 degree walkthrough VR video that demonstrates this interaction as an example.

(b) VR + Desktop Awareness (Bottom: VST): If each module works as a VST device, the user would have peripheral awareness of the real world while remaining focused on the central VR content. If the bottom modules works as VST, the user would have the desktop awareness to grab peripheral objects (e.g. a pen and note as shown in Fig. 3(b)) on the desk. Due to the potential limitations (e.g. delay, depth field mismatches, etc.) of VST [23], we believe that this VST view is not suited for full interaction but would be helpful for the HMD user to preview the surroundings to judge if clear vision is needed before removing the modules.

(c) VR + Peripheral Situational Awareness (Side: VST): In case the side modules turns to a VST, the user might be able to recognize a person approaching or sitting next to the user by looking out of the side modules while still paying attention to the central VR view (Fig. 3(c)). This configuration only provides unidirectional awareness to the HMD user, while bystanders in the real world have no idea what the HMD user is looking at.

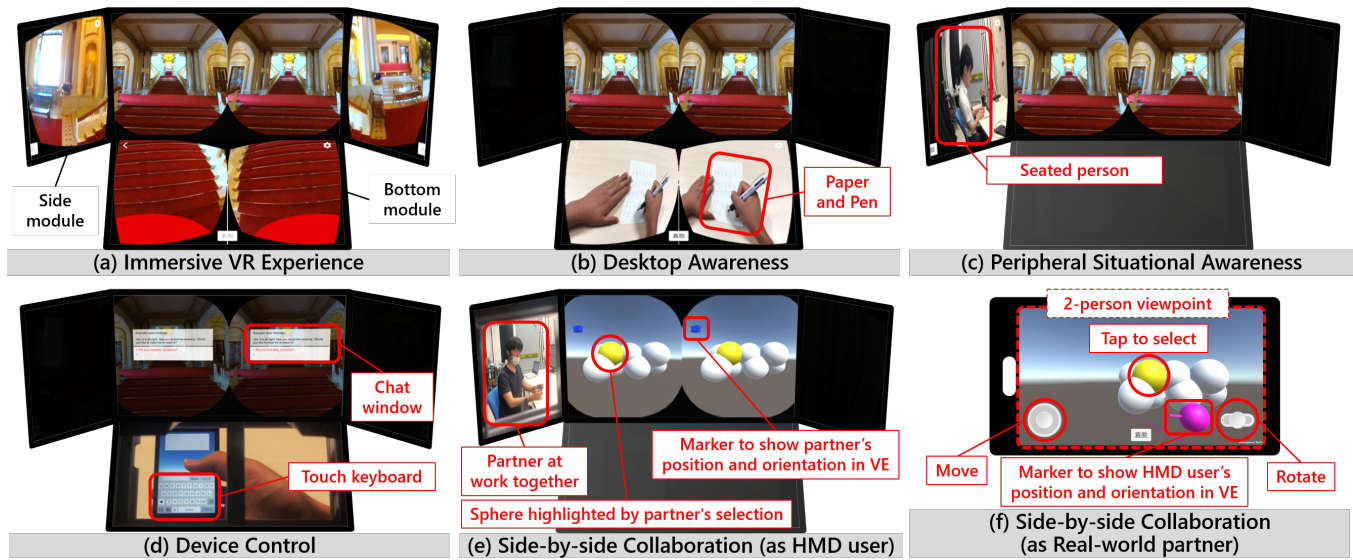


Figure 3: Interaction patterns using different HMD configurations: (a)-(e) show our prototype with the additional three modular views.

(d) VR + Device Control (Bottom: removed/reused): If the user removes modules, direct peripheral vision of real-world is enabled. If the bottom module is removed, the HMD user can directly see desktop devices (e.g. a keyboard or mouse, etc.) and his/her own hands through the opening, allowing easy device manipulations. The removed bottom module can also be reused immediately as an input device (e.g., keyboard like Fig. 3(d)), and the HMD user can operate it with fingers visible. This would be quite useful in mobile work contexts where voice inputs or VR hand controllers are not always available. Once the text typing is over, the module can be returned to restore the immersive VR experience.

(e, and f) VR + Side-by-Side Collaboration (Side: removed / reused): If the side module is removed, the empty space at the side would allow the HMD user and non-HMD users to casually understand the status (e.g., attention or gaze direction) of each other, providing them important cues and awareness for conversation. After initiating the conversation, they can immediately start asymmetric cross-device interaction with the removed side module being the interaction device for non-HMD users. Fig. 3(e) shows the view of the HMD user in an collaborative 3D modeling application (to change the color of virtual objects), in which a visual highlight is provided to indicate the pointer manipulated by the non-HMD user. Fig. 3(f) shows an example view of the non-HMD user's hand-held device passed by the HMD user. Compared to an ad-hoc connection using the non-HMD user's own smartphone, our approach is faster and more stable, offering an explicit protocol for initiating and ending the collaboration through the passing and returning actions. By returning the module, the HMD user can return to the occlusive VR experience.

3.3 Prototyping

3.3.1 Hardware design. We built a proof-of-concept prototype that comprises an Oculus Rift S as a sturdy base HMD, three display

modules, their holders, lenses, and attachments. To make room for the modules, we removed the original side and bottom cowls of Oculus Rift S. We used a 7th generation iPod Touch for each module because it is lightweight yet has adequate function with a high-resolution display, a backside camera, a long battery life, and wireless data communication. We used an Oculus Rift DK1 lens for the side and bottom modules.

We eventually decided to use a sliding mechanism for module removal and attachment like [25] because 1) its rail structure can be easily created via 3D printing without additional materials, 2) the module's movements are stable along with the created rails, and 3) it can be easily motorized in the future.

Fig. 4(a)-(d) shows how to build a side module. Except for the iPod Touch and the lens, all attachments and holders were 3D printed. To make sliding mechanism, the module holder has two rails. The assembled side module was affixed to the side of the headset belt with a printed attachment (Fig. 4(d)). We prepared two side modules, and each module weighs approximately 200 g. The side module can be removed by a two-step action. The HMD user first needs to slide the lens knob to the back to remove it from view (Fig. 4(e)). The user can then pull the module out from its holder by hand (see Fig. 4(f)). The removed module and lens can be restored by reversing these steps.

Fig. 5(a)-(d) shows how to build a bottom module. Similar to the side module, the module holder (right bottom in (a)) has two rails. Since the bottom module is seen by both eyes, two lenses were set in the module. The bottom module was also designed to be removed by a two-step action. Fig. 5(e) shows an HMD user changing their bottom vision by sliding the two lenses apart from each other, and Fig. 5 (f) shows that the user grasps the display module and pulls it up.

Fig. 6 on the left shows a fully assembled ModularHMD. The gap between the belt, modules, and base HMD is covered by shading film to ensure darkness inside the HMD. The total weight of a

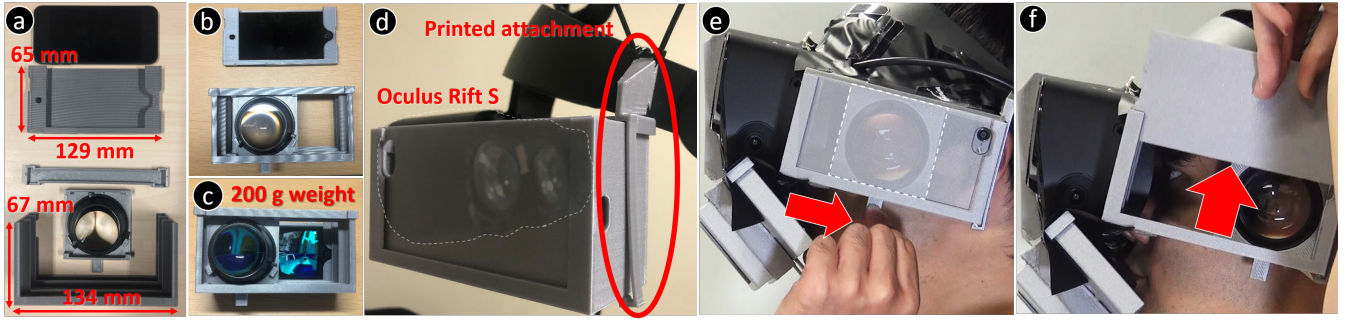


Figure 4: Side module: (a) all materials (an iPod Touch, its case, a lens with a knob, a module holder with two sliding rails), (b) an assembled display module and module holder, (c) a fully assembled side module, (d) a side module affixed to the side of the base HMD, (e) the slidable lens, (f) the display module being removed by pulling it up. The components enclosed by the dotted lines in (d) and (e) indicate important yet occluded parts of the hardware.

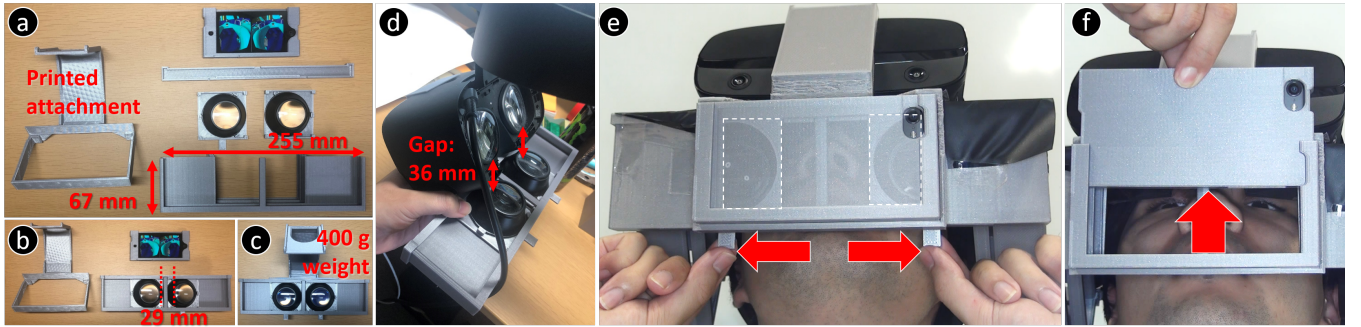


Figure 5: Bottom module: (a) all materials (an iPod Touch, its case, two lenses, a module holder, and an attachment), (b) An assembled display module, a module holder with two lenses, and an HMD attachment, (c) a fully assembled bottom module, (d) the bottom module fixed at the bottom of the base HMD cowl, (e) two slidable lenses, (f) the bottom module being removed by sliding it up. The components enclosed by the dotted lines indicate important yet occluded parts in (e).

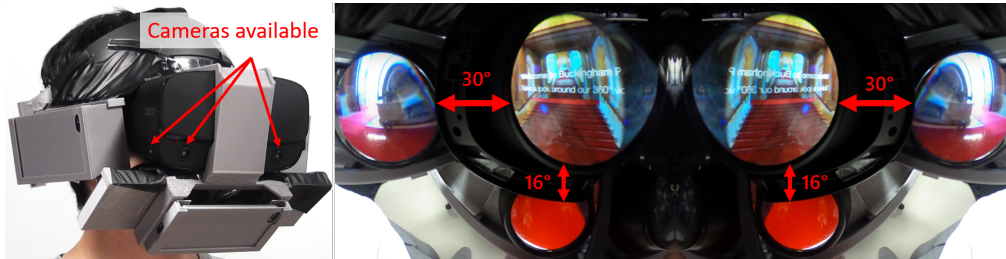


Figure 6: Prototype appearance: (left) A user wearing our ModularHMD prototype, (right) the interior structure (The left half of this image is mirrored to give the right half as it is difficult to take a wide-angle photo at this small distance.) of the prototype.

ModularHMD is 1.3 kg, including the base Oculus Rift S (500 g). Since three modules are installed around the front side of the HMD, the weight balance shifted to the front. Since this cannot be solved by fastening the head belt more tightly, we set a weight at the back for counterbalance. Fig. 6 right shows an inside view of ModularHMD. Four displays cover a wide FOV, but unfortunately we were unable to create a continuous VR view across the displays because of gaps between the lenses, shown in the figure. By contrast, we were able to create a stable and easy modular mechanism around the HMD.

3.3.2 Software. We created software based on a server-client architecture with TCP/IP communication. Using Unity 2019.4.7f1, we developed a Windows server software that drives the base Oculus Rift S HMD as well as iOS client software that drives each module. The module was set to one of the four display options; no (no content), virtual, VST, or reconfigured mode, depending on the given configurations. For example, in the case of virtual mode, the extended views were generated in Unity, where additional virtual side and bottom cameras were used, and then presented on each module display. In this implementation, we used explicit GUI button control

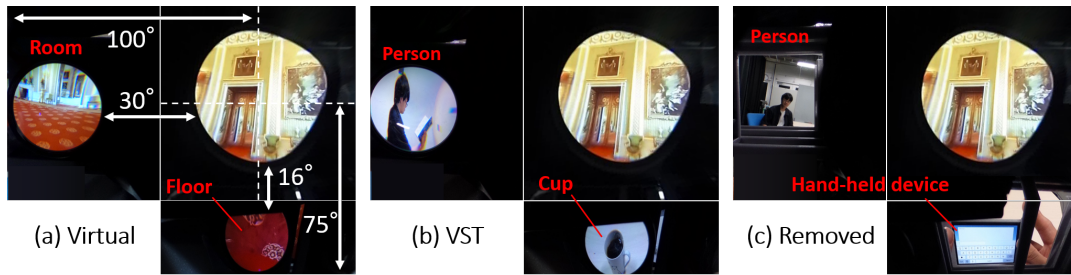


Figure 7: HMD user's monocular vision for each mode. Each figure is a cropped set of 360° cubemap images taken from the left eye's point of view with an omnidirectional camera, with the front, left, and bottom images cropped to fit a human's field of view.

to make the mode switching stable. As shown in Fig. 3(e)(f), We also created software to allow cross-device interaction between the base HMD (HMD user) and reconfigured input devices (partners or HMD users) (See Fig. 1(c)).

4 DISCUSSION AND FUTURE DIRECTION

4.1 Preliminary study

We conducted a preliminary qualitative user study to validate the ability of ModularHMD's each configuration/interaction, and ModularHMD's reconfiguration capabilities using previously mentioned interaction patterns. The feedback was generally positive regarding the capabilities of each configuration, especially HMD user's real-world awareness, device control, and ad-hoc collaboration with real-world partner. However, regarding the views of the peripheral lenses, we received opinions that the visibility of the peripheral lenses was poor due to the gap between the lenses (see Fig. 7 (a)(b)), and that VST had difficulty in visibility due to inconsistencies in depth perception. The proposed distributed workplace through module lending from the base HMD was also viewed positively. Most agreed that this makes sense as it can create an instant interaction device as well as an interaction window at the side of the HMD by simple action. However, some participants hoped that the two-step module slide operation could be done in one step.

4.2 Future work

We focused on a simplified module design, but there is a huge design space for expansion. For example, ModularHMD can be converted to FaceDisplay by changing each modular unit's face angle. We could also tackle the poor ventilation or high temperature issue [16] inside the HMD using modular mechanisms (e.g. module removal or use of a fan module).

There are numerous studies combining HMD and physical touchscreens or keyboards (e.g., [2]) in VR. The idea of ModularHMD would expand these interaction opportunities. For example, in temporal sketching in walkthroughs or dedicate touch input during volumetric data exploration, a rapid interaction switching between hands-free VR and VR with hand-held touchscreens would be highly helpful.

Once establishing the collaboration, the users' formation does not have to be side-by-side and it might be possible to create room-scale proxemic-based asymmetric interaction (like ShareVR [9]) by using the hand-held module as a see-thorough AR viewfinder.

Thus, the base HMD can be a powerful interaction platform for distributed workspaces to manage a large VR content.

Since our design concept was on peripheral modular devices with a base HMD, our insight can be directly transferred to current commercial HMD customization. Our results suggest re-purposing the side and bottom cowl of the HMD beyond the current occluding purpose. Our further insight is that lens/module arrangement would be challenging to provide a continuous view. To improve the visibility, we need to use specially designed Fresnel lenses [31] or redesign the modular structure. However, providing ad-hoc direct vision to the real-world (see Fig. 7 (c)) and quickly having instant interface device by simple sliding action might be an enough reason to consider integrating modular mechanics around the cowl or head-belt as accessories.

5 CONCLUSION

We proposed ModularHMD, a new HMD concept using a modular mechanism to support ad-hoc peripheral interaction with real-world objects and partners while maintaining an immersive VR experience for the HMD user. We designed and built a proof-of-concept prototype of ModularHMD using a base commercial HMD and three removable display modules installed in the periphery of the HMD cowl. The modules can be removed to ensure peripheral real-world awareness, reused as instant interaction devices (e.g. text keyboard), and then returned to terminate the peripheral interaction. We validated our concept through a preliminary qualitative user study, and the results suggested correctly configured ModularHMD helped HMD user's real-world awareness and device control, and ad-hoc collaboration with real-world partner. Future work includes modifying the total HMD structure to ensure continuous VR view, and a formal performance evaluation with custom HMDs.

ACKNOWLEDGMENTS

This work was partially supported by the Cooperative Research Project of the RIEC, Tohoku University.

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